KONE	TITLE: KDL16 Diagnostic Code List	DRAWING NO: **972485D01**
COMPILED BY: **HAT/L. Stolt**	PRODUCT: **KDL16**	ISSUE: **C**
CHANGED	CHECKED BY:	LANGUAGE: **en**
DATE **10.09.2009**	APPROVED BY:	SW: **MS EXCEL 2000**

A=supervision can be adjusted with parameters
Data 2 mode information: 1=normal, 2=inspection, 3=correction, 4=relevelling, 5=reduced speed

Main code	Sub code	Sub code name	Description	Data 1	Data 2	Reason	Detection	Operation	Recovery	Testing
102 Drive power stage										
102	**2001**	Motor overcurrent	Measured motor current too high.	0	0	Wrong motor parameters in 80 group, brakes do not open or mechanical problem hindering movement. Motor current measurement problem in MCDK.	Motor current is supervised by hardware and software overcurrent limit.	Stops immediately	Automatically	
102	**2002**	Drive overload	Measured drive load too high.	0	0	Wrong elevator parameters, mechanical problem hindering movement.	Drive output is supervised by software and if too much current with too low frequency is outputted supervision is tripped.	Stops immediately	Automatically	-
102	**2049**	Motorbridge internal error	MCDK fault logic tripped, but fault could not be identified.	0	0	Very short pulses in fault signals or broken circuitry.	DCBL tries to read fault from MCDK after its fault signal is set high and control pulses are removed but no fault can be read.	Stops immediately	Automatically	-
102	**2074**	Motorbridge IGBT	Fault detected with one of motorbridge IGBTs in MCDK.	0	0	Short circuit in motor. Broken IGBT.	MCDK motorbridge driver circuitry supervises correct operation of IGBTs.	Stops immediately	Automatically	-
102	**2075**	Braking IGBT	Fault detected with braking transistor in MCDK.	0	0	Broken IGBT.	MCDK braking transistor driver circuitry supervises correct operation of braking IGBT.	Stops immediately	Automatically	-
102	**2076**	Powerstage control voltage	Control voltage fault in MCDK	faulted voltage 24 = 24V	0	Voltage regulation malfunction in MCDK.	MCDK supervises 24V control voltage.	Stops immediately	Automatically	-
103 R **103**	**sistor** **1002**	**braking or regeneration** Braking resistor	Problem with braking resistor or transistor.	0	0	Braking resistor wrong type, not connected or broken. Braking transistor is conducting too long or is broken.	MCDK monitors braking transistor with hardware.	Drive is faulted.	Automatically	Disconnect braking resistor.
103	**2077**	Braking transistor overload	Too high peak or RMS current through braking transistor	0	0	Wrong resistor box size (too small Ohm) or too small drive module for application (too much power coming from motor when driving to light direction)	RMS current is calculated from motor power and dc link voltage. Peak current is calculated from rms current by dividing it with control duty.	Stops immediately	Automatically	-
103	**3069**	ECB-1 failed	ECB-1 failed state is detected	0	0	ECB-1 Internal error	Drive samples ECB-1 status signal and shows the warning message when the status signal blinks with 20% PWM of 5Hz.	Warning displayed.	Disconnect ECB-1	Unplug DC-link cable from ECB-1
104 Motor protection										
104	**2004 A**	Motor temperature	Motor thermistor (PTC) or NTCsensor indicates overtemperature.	0	0	Motor overheated, thermistor/NTC broken or thermistor/NTC cable disconnected or wrong wiring in XT1 connector.	Drive measures voltage over thermistor (PTC) or temperature with NTC. With NTC, supervision limit can be adjusted with parameter.	Drive is faulted at floor.	Automatically	Disconnect thermistor/NTC cable and/or replace thermistor/NTC with potentiometer and adjust resistance to trip supervision.
104	**3057 A**	Motor temperature warning	Motor thermistor or NTC-sensor indicates overtemperature.	0	0	Motor overheated, thermistor/NTC broken or thermistor/NTC cable disconnected or wrong wiring in XT1 connector.	Drive measures voltage over thermistor (PTC) or temperature with NTC. With NTC, supervision limit can be adjusted with parameter.	Drive is faulted at stop.	Automatically	Disconnect thermistor/NTC cable and/or replace thermistor/NTC with potentiometer and adjust resistance to trip supervision.
105 Power supply										
105	**2006**	DC bus voltage	DC bus voltage is out of operational limits.	1=undervoltage 2=overvoltage	0	Wrong braking resistor. Braking resistor or transistor not working, wrong system parameters.	MCDK measures DC bus voltage and has hardware overvoltage limit 800V. Drive supervises DC bus undervoltage by software.	Stops immediately	Automatically after DC bus voltage is within normal limits.	Replace braking resistor with higher ohmic and run elevator to generative direction.

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106 Drive										
106	**1001 A**	TFC (Tacho Fault Counter)	Tacho fault counter has exceeded set limit.	0	0	Motor encoder broken, problem with encoder cables or wrong system parametrization.	Drive monitors speed difference between motor sensors if error is detected tacho fault counter is increased. Counter is resetted after successful run.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	Disconnect motor encoder from drive and try to run several times.
106	**1055**	Software version	Software update has failed	0	0	Incompatible software versions in two processors	If version do not match supervision is tripped.	Drive is faulted.	By user: update new software from SD card	Update TI2802 with different software version than M16C.
106	**1056**	Movement risk during start or stop sequence	Sequence control has detected failure which could cause unintended movement	0	0	Brake close time is too short. Problem in BCR relay control. Wiring mistake.	Motor movent is monitored from encoder. Brake current is monitored during start/stop sequence.	Drive if faulted.	By user. Switch to inspection/RDF mode or cycle power.	Using modified version of BCR or by causing movement during start sequence.

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107 Load weighing										
107	**2008**	Load weighing	Load weighing sensor signal out of range or levelling tried without LWD setup.	0	0	Broken sensor, problem with sensor cabling or missing LWD setup.	Drive measures sensor current and if value is too low or high supervision is tripped. Levelling is not allowed without LWD setup.	Drive will not start.	Automatically when signal is in range or LWD setup is done.	Disconnect LWD from drive.
107	**3020**	Load weighing warning.	LWD setup not done or calculated load out of range.	0	0	Bad/missing LWD setup or misplaced sensor.	Drive calculates load based on given setup and if value is too low or high warning is given.	Normal operation is allowed but warning is displayed after run.	Automatically when calculated load is in range or LWD setup is done.	Make LWD setup with wrong values.
108 T	**rque/Speed controller**									
108	**2009**	Speed difference	Difference limit between speed reference and measured motor speed exceeded.	position [cm]	speed+mode [cm/s] + last number is mode	Torque feedforwards are not tuned, speed controller is not tuned, problem with motor speed feedback or wrong system parametrization.	Drive compares speed reference and motor speed feedback and if difference is too big supervision is tripped.	Stops immediately.	Automatically	Disconnect motor encoder during operation.
108	**2013**	DTC (Dual Tacho Control)	Difference limit between speed feedback and motor synchronous frequency exceeded.	position [cm]	speed+mode [cm/s] + last number is mode	Motor encoder broken, problem with encoder cables or wrong system parametrization.	Drive compares speed feedback to output voltage frequency when active power is not close to zero. If difference is too big between them supervision is tripped. Tacho fault counter is increased except in inspection/rescue drive modes.	Stops immediately or ramp stop to next floor.	Automatically	Disconnect motor encoder during operation.
108	**3004 A**	Torque limit warning	Torque output of speed controller is at limit.	0	0	Too weak drive for application or wrong system parametrization.	Drive supervises speed controller output and if that is saturated too long supervision is tripped.	Warning displayed.	Automatically	Set high acceleration value and use heavy load.
108	**3062**	Drive output voltage warning	Drive output voltage is at limit.	0	0	Supply network voltage is too low or wrong system parametrization.	Drive supervises current controller output and if that is saturated too long supervision is tripped.	Warning displayed.	Automatically	Lower supply network voltage.
109 P	**sitioni**	**ng**								
109	**3065**	Car position 77 error	Drive has lost accurate position and needs a synchronisation run.	floor	0	Problem with 77 signals, motor encoder broken, problem with encoder cables or wrong system parametrization.	Drive compares current and stored 77 positions and if difference is too big supervision is tripped.	Ramp stop to next floor.	Automatically after position information is consistent and position is verified.	Set 77U/N on at the middle of shaft.
109	**3066**	Car position 61 error	Drive has lost accurate position and needs a synchronisation run.	floor	0	Problem with 61signals, motor encoder broken, problem with encoder cables or wrong system parametrization.	Drive compares current and stored 61 positions and if difference is too big supervision is tripped.	Ramp stop to next floor.	Automatically after position information is consistent and position is verified.	Set 61U/N on at the middle of floors.
110 Drive temperature										
110	**2014**	Heatsink temperature	Heatsink too hot (about 75C in normal mode or about 90C in inspection mode)	0	0	Excessive load on drive, drive fan is not operating, ambient temperature is too high.	Drive measures heatsink temperature.	Drive is faulted at floor.	Automatically when heatsink temperature is low enough.	Heat heatsink.
110	**3010**	Heatsink temperature warning	Heatsink too hot (about 75C)	0	0	Excessive load on drive, drive fan is not operating, ambient temperature is too high.	Drive measures heatsink temperature.	Only inspection run is possible.	Automatically when heatsink temperature is low enough.	Heat heatsink.
110	**3011**	Control board temperature warning	DCBL temperature is over warning level (about 65C).	temperature [°C]	0	Ambient temperature too high.	Drive measures board temperature and if value is over limit warning is given.	Warning displayed.	Automatically when board temperature is low enough.	Heat DCBL board.

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118 Shaft setup										
118	**1043**	Shaft setup missing	No valid shaft setup.	0	0	Shaft setup is not done or existing setup has been deleted.	Drive monitors setup data and if data is invalid supervision is tripped.	Drive is faulted. Only setup or RDF/inspection drive is allowed.	By user: switch to inspection/RDF mode.	Delete existing setup by changing traction sheave diameter and switch to automatic use.
118	**2021**	Shaft setup start position	Setup started too close to 61:N vane or 61:U missing.	0	0	Car is not well below 61:N edge before start or problem with 61 signals.	Drive monitors 61 signals during setup and if 61:N is active or 61:U is missing at start supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Start setup very near of 61:N edge.
118	**2022**	Shaft setup 77:S	77:S edge detected in wrong place. 77:S must be inside 77:U/N zones.	0	0	Problem with 77:U/N/S signal.	Drive monitors 77:U/N/S signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Cause wrong 77:S signal during setup.
118	**2023**	Shaft setup 77:U/N	77:U/N edge detected in wrong place. N must be at bottom, U at top, U and N cannot be active at the same time and 77:S must be inside 77:U/N zones.	0	0	Problem with 77:U/N/S signal.	Drive monitors 77:U/N/S signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Cause wrong 77:U/N signal during setup.
118	**2025**	Shaft setup 61:N sequence	61:N edge detected in wrong place. 61:N must come on when 61:U is on and must go off when 61:U is off.	0	0	Problem with 61:U/N signals.	Drive monitors 61:U/N signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Disturb 61 signals during setup
118	**2026**	Shaft setup 61:U sequence	61:U edge detected in wrong place. 61:U must come on when 61:N is off and must go off when 61:N is on.	0	0	Problem with 61:U/N signals.	Drive monitors 61:U/N signals during setup and if sequence is invalid supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Disturb 61 signals during setup

118	**2027**	Shaft setup maximum floor count	Maximum floor count exceeded. (max. 128 floors)	0	0	Problem with system configuration.	Drive records floor information during setup and if maximum floor count is exceeded supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	Make shaft setup with too many floors. (simulator)
118	**2029**	Shaft setup position count	Position is not increasing normally.	0	0	Problem with motor encoder or its cabling.	Drive calculates position from encoder during setup and if position is not increasing normally supervision is tripped.	Stops immediately	By user: switch to inspection/RDF mode.	
118	**2073**	NTS setup	Wrong NTS switch order.	0	0	Problem with NTS switches: 77U, 77N	Drive supervises NTS switches during setup and if the switches are detected in wrong order supervision is tripped.	Stops immediately		
118	**6010**	Shaft setup minimum overlap info	Reports minimum 61:U/N overlap	floor	overlap [mm]	Too small overlap can be caused by problems with 61 signals.	Drive measures accurately every edge of 61 signals and calculates minimum overlap.	Value is shown after setup	-	Cause very short and very long overlaps
118	**6011**	Shaft setup maximum overlap info	Reports maximum 61:U/N overlap	floor	overlap [mm]	Too large overlap can be caused by problems with 61 signals.	Drive measures accurately every edge of 61 signals and calculates maximum overlap.	Value is shown after setup	-	Cause very short and very long overlaps
120 Start sequence										
120	**1025**	Wrong Motor Current	No Motor current or one phase current missing	0	0	Problem in motor wiring	By motor controller	Drive is faulted	Start is prevented until problem is fixed	By disconnecting one motor phase.
120	**1037**	Motor bridge	Motor bridge does not start	0	0	MCDK broken.	Drive tries several times to start and if motor bridge does not start supervision is tripped.	Drive is faulted.	By user: switch to inspection/RDF mode or cycle power.	-
120	**2020**	Safety relay	Safety relay input does not activate in start or is stuck after run.	0	0	Safety relay wiring disconnected or broken, auxiliary contact for safety relay broken. Wiring shortcircuited.	Drive monitors safety relay input by software.	Drive is faulted.	Automatically when safety relay input works.	Disconnect or short safety relay input.
120	**3070**	Movement during angle detection	Rotor movement detected during angle detection. Results might be wrong and start may fail.	0	0	Wrong brake adjustment. Too weak brakes.	Drive monitors encoder movement during angle detection	Warning displayed.	Automatically	Manually weaken brakes.
123 Motor setup										
123	**1007**	Motor setup fault	Motor parameters not valid.	0	0	Motor parameters (6_80 through 6_84) are wrong.	Drive software monitors parameters.	Drive is faulted.	By user: enter valid motor parameters.	Set a motor parameter to zero and switch elevator to normal mode.

Maincode	Sub code	Sub code name	Description	Data 1	Data 2	Reason	Detection	Operation	Recovery	Testing
124 Drive feedbacks										
124	**1012**	77 switch fault	77:U/N active at the same time or 77:U/N active at wrong place.	0	0	Broken/malfunctioning 77 switch or problem with switch cabling.	Drive monitors 77 switch information and if both 77:U and 77:N are active at the same time or 77:U/N is active at wrong place supervision is tripped.	Drive is faulted.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Set both 77 switches on at the same time.
124	**1013**	77S switch fault	77S stuck	0	0	Broken/malfunctioning 77S switch or problem with switch cabling.	Drive monitors 77S and if it is on elsewhere than terminal floors supervision is tripped.	Drive is faulted at floor.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Set 77S on continuously.
124	**1032**	77S switch fault	77S missing	0	0	Broken/malfunctioning 77S switch or problem with switch cabling.	Drive monitors 77S and if it is not on in terminal floors supervision is tripped.	Drive is faulted at floor.	Automatically when switch information is correct or by user: switch to RDF/inspection mode.	Disable 77S switch.
124	**1049**	Heatsink temperature sensor	Heatsink temperature sensor faulted.	0	0	Heatsink temperature sensor not connected, broken or problem with sensor wiring.	Drive supervises heatsink temperature sensor signal.	Drive is faulted.	Automatic	Disconnect heatsink sensor from MCDK.
124	**1054**	SD card fault	Normal mode is not allowed if SD card is in the reader.	0	0	SD card can only be used for sw update and must be removed after the update is ready.	Drive supervises SD card socket.	Only inspection run is possible.	Automatic, after SD card is removed from reader.	Insert SD card and set elevator to normal mode.
124	**2057**	Motor encoder	Motor encoder faulted.	0	0	Motor encoder not connected, broken or problem with sensor wiring.	Drive supervises motor encoder wiring and calculated speed.	Drive is faulted at floor.	Automatic	Disconnect encoder.
124	**3060**	NTS switch position	Detected NTS switch position does not match with setup.	position [cm]	speed [cm/s]	Problem with motor encoder, bad NTS setup, problem with NTS switches or wrong system parametrization.	NTS function compares detected switch positions between saved positions and in case of too large error supervision is tripped.	Ramp stop to next floor.	Automatic	Move NTS switch(es).
124	**3061**	NTS switch order	Wrong NTS switch order.	position [cm]	speed [cm/s]	Problem with NTS switches: 77U, 77U:1, 77U:2, 77U:3, 77U:4 77N, 77N:1, 77N:2, 77N:3, 77N:4	NTS function supervises that switches appear in correct sequence. In case of error supervision is tripped.	Ramp stop to next floor.	Automatic	Swap NTS switches.
124	**3064**	SD card fault	Drive cannot read SD card.	2=missing files 3=checksum 4=SD card type	0	SD card is corrupted, empty or does not have correct files.	Drive identifies SD card when it is inserted to reader.	SD card cannot be used for sw update.	Automatic	Insert empty SD card.
124	**3068**	Ramp slowdown due ommunication supervision	Communication link to the LCECPU has failed for moment	position [cm]	speed+mode [cm/s] + last number is mode	Disturbance, grounding error, broken wire, control system failure	Communication link is checked continuously, timeout is 500ms	Ramp stop to next floor if main contactor stays on.	Automatic	Cause disturbance to the communication link
126 Mechanical brake										
126	**1045 A**	Brake current with reduced voltage	Detected brake current with reduced voltage failed too many times.	0	0	Problem with brakes or with brake control module.	This fault occurs if there's too many consecutive 126_3059 warnings.	Drive is faulted.	By user: switch to inspection/RDF mode.	
126	**1046**	Automatic brake test	Automatic brake test has failed too many times.	0	0	Problem with brakes.	This fault occurs if there's too many consecutive 126_2071 or 126_2072 faults.	Drive is faulted.	By user: switch to inspection/RDF mode.	
126	**1047**	Brake current	No brake current too many times.	0	0	Problem with brakes or with brake control module.	This fault occurs if there's too many consecutive 126_2104 faults.	Drive is faulted.	By user: switch to inspection/RDF mode.	Disconnect brake cable from brake control module and try to drive.
126	**2071 A**	Brake test failed: speed	Speed detected from motor.	0	0	Problem with brakes.	Drive supervises speed during automatic brake test. This supervision is active only if automatic brake test is selected in use.	Drive is faulted.	Automatic	During brake test rotate motor or motor speed sensor.
126	**2072 A**	Brake test failed: position	Movement detected from car.	0	0	Problem with brakes.	Drive supervises elevator position during automatic brake test. This supervision is active only if automatic brake test is selected in use.	Drive is faulted.	Automatic	During brake test move elevator car.
126	**2104 A**	Brake current	No brake current detected when opening brakes.	0	0	Problem with brakes or with brake control module.	Drive monitors brake current and if current is not detected supervision is tripped.	Drive is faulted.	Automatic	Disconnect brake cable from brake control module and try to drive.
126	**3059 A**	Brake current with reduced voltage	Detected brake current with reduced voltage is too high.	0	0	Problem with brakes or with brake control module.	Drive monitors reduced voltage brake current and if current will not be smaller than with full voltage supervision is tripped. Supervision is active only when reduced voltage is selected in use.	Drive is faulted.	Automatic	
126	**6021**	Brake 1 test passed	Reports success of brake 1 manual test.	0	0	User har performed manual brake 1 test.	Drive determines whether test was successfull or not.	Code is shown after successful test.	-	Perform manual brake test.
126	**6022**	Brake 2 test passed	Reports success of brake 2 manual test.	0	0	User har performed manual brake 2 test.	Drive determines whether test was successfull or not.	Code is shown after successful test.	-	Perform manual brake test.

End of table.